

Instructions: Please answer the following questions making sure to write your answers legibly on the second page. The code below is a segment of a Python Flask application that provides context for the questions below.

- Since we are writing Python code, please be careful about spacing/indentation. Leave enough room that it is obvious when you want things indented.
- Make sure to write legibly.
- All questions are worth equal points unless otherwise stated.
- The quiz is cumulative and may include material from previous weeks.

```
from flask import (
    Flask,
    request,
    Response,
)

from data_processor import (
    is_night,
    logger
)

app = Flask(__name__)

# WHERE YOUR ANSWER WILL BE

@app.route('/api/v1/data', methods=['GET'])
def return_data_from_api_interface():
    '''
    Main Data route for our flask app
    '''
    data = 'Hello World!'
    logger.info("REQUEST: api/v1/data endpoint")
    return Response(data, status=200,
                    headers={'Content-Type': 'text/html'})

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

We also have the following Makefile. Note that the Docker image, via a `CMD` statement, starts the Flask app as the default command.

```
IMAGE_NAME=flask_app

.PHONY: build run_flask

build:
    docker build . -t $(IMAGE_NAME)

run_flask: build
    docker run -p 4000:5000 $(IMAGE_NAME)
```

1. Our Flask server puts logs in the following location `/logs/flask.log` (on the host not inside the container). The command `logger.info()` writes the contents of its string argument to that log file. Each call writes a new line to the log file. You can find an example of how it is applied in the `/api/v1/data` route above.
 - (a) Write a Make command that returns the number of times that the `api/v1/data` endpoint is called (you can ignore the `PHONY`).
 - (b) Write another Make command that shows the newest entries (i.e., the last 10 lines in the file) from the log (you can ignore the `PHONY`).
2. We want to write a route (`/api/v1/dark_out`) which responds to a `GET` request and which tells us if it is dark out. The imported function `is_night` returns `True` (boolean) if it is nighttime, `False` (boolean) if it is daytime, and `None` if it is dusk or dawn.
 - Please write a complete route (including all required decorators, etc.) that can be placed into the file where indicated.
 - It should return status code 200 if it is nighttime (`True`)
 - It should return status code 201 if it is daytime (`False`)
 - It should return status code 202 if it is dusk or dawn (`None`)
 - The body and headers can be empty.
 - No need for comments or docstrings.